

Exercises Logical Operation

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1. Given a 16-bit value x, use a combination of shift operations and bitwise logical operation to obtain the result of exchanging the upper 12 bits and lower 4 bits in the variable y.

Misal:

$x = 0x1234$ // 0001 0010 0011 0100

Hasil akhirnya mau gini:

$y = 0x4123$ // 0100 0001 0010 0011

Caranya gimana?

$y1 = x \gg 4 = 0000\ 0001\ 0010\ 0011$

$y2 = x \ll 12 = 0100\ 0000\ 0000\ 0000$

$y = y1 \mid y2 = 0100\ 0001\ 0010\ 0011$

Jadi jawabannya:

$y = ((x \gg 4) \mid (x \ll 12)) \& 0xFFFF$

2. Given a 16-bit value x, use a combination of shift operation and bitwise logical operations to obtain the value of the 7th bit from the lowest bit (the LSB is counted as the 1st bit) and saving it into the variable y.

Misal:

$x = 0001\ 0010\ 0100\ 0001$

Berarti $y = 1$

y merupakan value dari bit 7 dimana LSB dihitung sebagai bit 1

Cara:

$y = x \gg 6 = 0000\ 0000\ 0100\ 1001$

Jadi jawabannya:

$y = (x \gg 6) \& 1$